



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 02- 306 - 1

Date: 3-08-2021

EDMS 02-306
SPECIFICATION FOR
Enclosures for Energy Meters Grouping
Single / Three Phase Direct Meters
With Indirect Master Meter

Issue: Aug. 2021 / Rev. 1



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- Option items should be specified byEDC before tendering.
- Latest edition of EDMS 17-300 (Low Voltage Current Transformers) should be attached and transformer ratio should be defined.



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1. Scope

This specification covers the minimum requirements for design, manufacturing, testing, and supplying of Enclosure for low voltage Energy Meters Grouping Single / Three Phase Direct Meters with Indirect Master Meter. The enclosure should be provided with places/spaces for energy meters, indirect master meter and equipped with main circuit breaker, miniature circuit breakers, bus bars, and current transformers. The enclosure should be effectively grounded.

2. Applicable Standards

Unless otherwise specified in this specification, the enclosure should comply with the latest edition of the following standards. The enclosure should be designed, manufactured, and tested in accordance with the applicable standards. In case of any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

No.	Standard	Description
1	IEC 60529	Classification of Degree of protection
2	IEC 61439-1	Low Voltage Switchgear and Control Gear-General Rules
3	IEC 60114	Indicating Instruments
4	IEC 62208	Empty Enclosures For Low Voltage Switchgear and Control Gear
5	IEC 60085-2007	Electrical insulation-thermal evaluation and designation
6	IEC60898-1	Circuit-breakers for overcurrent protection for household and similar installations –Part 1 : Circuit-breakers for A.C operation
7	EDMS 02-300-2	Low Voltage Distribution Panels

3. Environmental Condition

The required enclosures should be suitable for satisfactory continuous operation under the following:

	Condition	Specification
1	Maximum ambient temperature (°C)	45 °C (50°C - as option)
2	Minimum ambient temperature (°C)	-5
3	Maximum relative humidity (%)	95
4	Maximum altitude	1000 m



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4. General Requirements

- 1- The enclosure should be made of hot rolled steel intended to contain energy meters for wall-mounted installation (exposed or flush) or with a frame base suitable for the free-stand enclosure or polycarbonate with thickness 6 mm according to....EDC requirements.
- 2- The enclosure should be manufactured with appropriate dimensions according to: the required size of the apparatuses installed inside it, the technical industrial principles and energy meter types (smart/prepaid/three-phases/single-phase).
- 3- The window dimension should be appropriate for single /three-phase energy meter of any type by half bunched for three and single phase.
- 4- The enclosure should be suitable to group (4, 6, 8 or 12) energy meters in addition to master meter according to the requirements of EDC after the technical site survey.
- 5- The enclosure should have an adequate mechanical strength to withstand rough handling as may be expected in normal uses.
- 6- The manufacturer should carry out all internal wiring (from the main bus bar to the energy meter terminals).
- 7- The connection shall not be braided or soldered. Wiring shall be identified by colors and numbers (red, yellow and blue for phases, black for neutral) - installed inside a suitable duct.
- 8- The enclosure should be equipped with external antenna or the material of the enclosure meter should not obstruct wireless communication operation in case of smart meters design model.



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5. Design Criteria :

1. The enclosure should be made from hot rolled steel thickness not less than 2 mm before painting with electrostatic painting thickness not less than 60 micron with a light gray color, smooth outer finish or polycarbonate with thickness 6 mm according to....EDC requirements.
2. . The enclosure should have holes from bottom with a suitable diameter for the incoming and outgoing cables (depending on the rated load of the enclosure and the number of the energy meters) with rubber gland to seal the holes to prevent the entry of dust or leakage of objects inside the enclosure.
3. The enclosure body and hinged doors should be grounded by insulated flexible copper wire with a cross-section area at least 6mm (yellow-green PE).
4. Protection degree of the enclosure $\geq$ IP41 with a transparent movable door (articulated) in the main door/mirror of the enclosure for each breaker.
5. In case of outdoor installation, the enclosure should have the following / characteristics:
 - a) IP 54
 - b) The panel should be provided with a sloping roof to prevent rain accumulation (slope angle should not be less than 6° horizontally)
 - c) Enclosure should be made from sheet steel galvanized with thickness should be not less than 2 mm before electrostatic painting or poly carbonate with thickness not less than 6 mm.
 - d) The outside door should be without any window.
6. The enclosure should be suitable for wall mounted or free-stand installation.
7. Numbering (Addressing) of each energy meter (Labels).

8. Enclosure ratings:

Enclosure	MCCB rating	MCB'S rating for each meter	Cross section area of internal conductor copper	Cross section area of main Bus bars copper
4 – 1PH	200 A	80 A	16mm ² ,cu	125 mm ²
6 – 1PH	200 A	80 A	16mm ² ,cu	125 mm ²
8 – 1PH	300 A	80 A	16mm ² ,cu	200 mm ²
12 – 1PH	400 A	80 A	16mm ² ,cu	250 mm ²
4 – 3PH	500 A	100 A	35mm ² ,cu	300 mm ²



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6 – 3PH	800 A	100 A	35mm ² ,cu	400 mm ²
8 – 3PH	1000 A	100 A	35mm ² ,cu	600 mm ²
12 – 3PH	1500 A	100 A	35mm ² ,cu	800 mm ²

6. Enclosure Components

1. The enclosure should be contain:

1. Space for Master Energy Meter (electronic -5A- 3*220/380 V) to registers the total consumption of the building, and detects "Enclosure door open", any tamper then record the event/tampers.
 2. Space for energy meters.
 3. Main circuit breaker M.C.C.B according to specs.EDMS19-303-1 and approved by EEHC.
 4. Current transformers according to specs.EDMS 17-300-1 and approved by EEHC with suitable clearance between C.Ts during installation.
 5. Main bus bar with current density not more than 1.5 A/mm² with a purity of at least 99.9% (conductivity 57 SM/m), and insulated by heat shrinkable tube.
 6. Miniature C.B for each energy meter:
 - In the case of prepaid meter: M.C.B should be connected before each meter and MCB switching lever accessible from outside the enclosure without opening the inner door it/mirror.
 - In the case of smart meter: M.C.B should be connected before each meter and MCB switching lever accessible only from inside the enclosure after opening the inner door it/mirror by...EDC.
 7. All required wiring, the cross section area of conductor should be not less than 16mm (copper).
 8. Tampering indication led controlled by the master meter.
 9. Alarm reset button.
 10. Indication lamps (heavy duty led), limit switch , mute button and horn alarm that operates when the indoor/mirror panel is opened illegally without the master key / without the knowledge of the electricity company and can be reset with push-button.
 11. External antenna in case of smart meter.
2. The cross-section area of the neutral bus bar should be at least 60% of the CSA of the phase's bus bars.
 3. The cross-section area of the earthing bus bar should be at least 25% of the CSA of the phase's bus bars.



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4. The bus bars should be mounted on insulators made from porcelain or cast resin.
5. The main, neutral and earthing bus-bars should have holes and equipped with galvanized steel bolts and washers necessary for connecting cables.
6. All the bus-bars should be insulated with " heat shrinkable" colored in red, yellow, blue, and black for the three phase, and neutral bus-bars, respectively.

7. Safety Conditions of The Enclosure (4, 6, 8 or 12) Energy Meters

7.1 General

1. Manufacturers should provide all means of safety to prevent illegal works of unauthorized persons such as energy meter tampering or energy theft.
2. The inner door of the enclosure has appropriate place for sealing and the energy meters should be sealing too.
3. The enclosure should have the facility to be secured against tampering from unauthorized persons such as energy meter tampering or energy theft by sealing the inner door/mirror and provided with limit switch to indicate any tampering with the energy meters when the panel opened.
4. The inner door (mirror) should be provided with:
 - a) Limit switch to indicate any tampering with the energy meters when the panel opened.
 - b) Secured with a (master key) to restrict the access to authorized persons only from the ...EDC.
 - c) Suitable window to allow meter reading, charging operations, screen scrolling button, and MCB switching lever from outside the enclosure without opening the inner door it/mirror.
5. The customers will have the possibility to open the outside door of the enclosure to deal with the energy meters, charge them and follow their consumption at any time; the outside door will be secured using two regular locks that can be opened with any key.
6. The outside door should be with a damper.

7.2 In case of Enclosure for (8 or 12) meters:

When the meter enclosure consists of two hinged doors, door with window for customer meters and another door for (the master meter, the main circuit breaker, and B.B) the following shall be considered:



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1. The door with window for customer energy meters should be secured by a limit switch to restrict access to authorized persons, where it can't be opened without opening the main door of the master meter
2. The door with window should be provided with a (limit switch) to operate the horn alarm with a reset push button inside the enclosure to reset the horn alarm only without disconnecting the energy meters and tampering indication led controlled by the master meter.

8. Routine and acceptance tests:

1. ...EDC reserves its rights to witness the routine testing of the energy meter enclosure.
2. Attendance of (...EDC) inspection engineers during testing should not relieve the manufacturer from his full responsibility for manufacturing the energy meter enclosure under the requirements of this specification and should not give him the right to invalidate any claim that (... EDC) may rise because of defective or unsatisfactory material or faulty workmanship.
3. Routine tests of all components should be carried out according to the latest IEC standards by the manufacturer at his lab or any certified and approved lab and on his charge.
4. Visual inspection, dimensions measurement, thickness of sheet, and painting.
5. Phase's insulation test.
6. Operation test.

9. Marking

The following information should be clearly and indelibly embossed (engraved or laser print) on the cover of the enclosure.

1. Manufacturer's name.
2. Year of manufacture.
3. Danger of singe plate.
4. A printed warning in Arabic "ممنوع فتح الباب الداخلي للوحة ومن يخالف ذلك يتعرض للغرامة".
5. Purchase order number and date.



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6. Logo of the electricity company.

10. Inspection

1. The representatives of the ...EDC have the right to visually inspect the quantity supplied up to 100% including components.
2. Operation test and insulation tests should be applied for 10% at least of the order and may be increased according to ...EDC requirements.

11. Submittals

1. Attached drawings (Drawings1-Drawings2) are only for general guidelines, however, the detailed dimensional drawing showing clearly the dimensions and material and its constructional features should invariably be furnished with the offer.
2. Approval certificate from EEHC.

12. Guarantee

Guarantee: Not less than 12 months from the date of installation or 18 months from the date of delivery.

Guarantee Table for panel type 4x/6x/8x/12x- with a master meter

<u>No</u>	<u>Item</u>	<u>Vendor Specs.</u>
General:		
1	IEC Standards	
2	Manufacturer's name	
3	Number of energy meters	
Enclosure:		
1	Material	
2	Height	mm
3	Width	mm
4	Depth	mm



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5	Thickness	mm
6	Color	
7	IP degree	
8	Holes of cables	Bottom/side
9	Size of viewing window	mm x.....mm
Bus bars		
1	Size of bus bar	mm x.....mm
2	Size of Neutral bus bar	mm x.....mm
3	Conductivity of bus bars	MS/m
Accessories:		
1	Main C.B. type and its rate	
2	Branches C.B. type and its rate	
3	C.S.A of internal wiring	mm
Safety:		
1	Matching all requirements for safety conditions of the enclosure meters	YES/NO
Marking:		
1	Matching all requirements for Marking	
Submittals:		
1	All drawings	YES/NO
2	Samples	YES/NO
Guarantee:		
1	Guarantee period	

We guarantee the data given above for the equipment offered.

Signature:

Date:



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Recommendations for ...EDC

The number of the enclosures in the residential building should be determined according to the number of apartment in it:

- a) In the case of the building that contains 12 apartments or less, one enclosure should be installed and whose location should be determined by the distribution company at the entrance of the building in an architecturally acceptable area.
 - The enclosure should be suitable to include the master meter, main circuit breaker, the bars, the current transformers and the meters of the customers in one enclosure up to 12 energy meters.
- b) In the case of the building that contains more than 12 apartment:
 - An enclosure should be installed every three floors or each floor in which the energy meters for these floors are placed with master meter.
 - A main door/mirror of the enclosure should be installed with a window per energy meter and each window has a transparent movable door (articulated) in the main door/mirror of the enclosure.
 - The master meter only should be installed in one enclosure with main circuit breaker, current transformer and main bus bar to register the total consumption of the building installed at the point of common coupling.
- c) The enclosure ratings according to the rated maximum current of the meter but The Electricity Distribution Company responsible to determine the rated current of the main circuit breaker (MCCB), miniature-circuit breakers (MCB'S) and type of the energy meters before tendering according to the load of the building. Preferable to do site survey to choose the correct enclosure size.

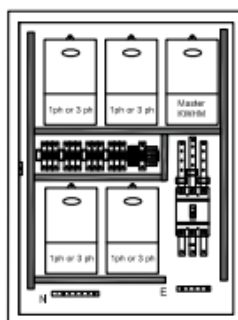


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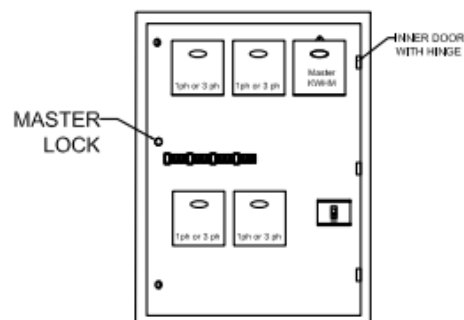
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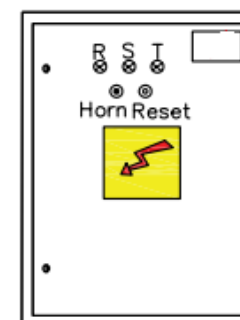
13.Drawings-1



FRONT VIEW
(INSIDE)



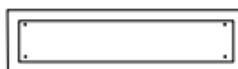
FRONT VIEW
(With Mirror)



FRONT VIEW
(OUTSIDE)



SIDE VIEW



Down View

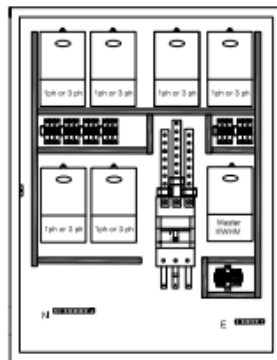
Enclosure for meters grouping V4



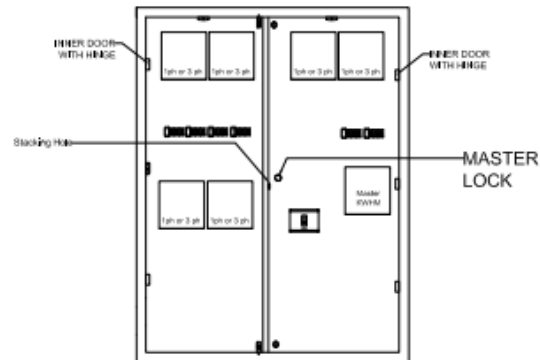
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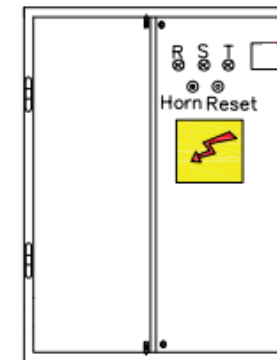
Date: 3-08-2021



FRONT VIEW
(INSIDE)



FRONT VIEW
(With Mirror)



FRONT VIEW
(OUTSIDE)



SIDE VIEW



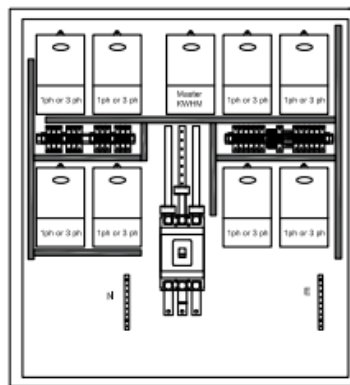
Down View

Enclosure for meters grouping V6

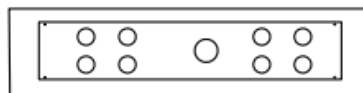
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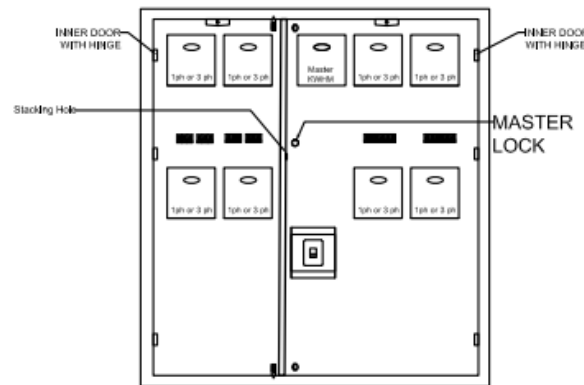
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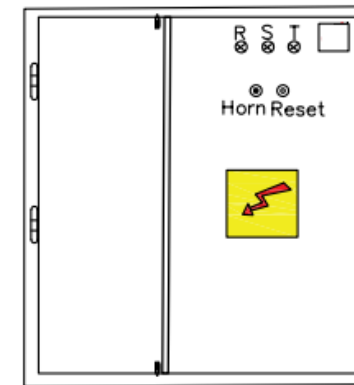
FRONT VIEW
(INSIDE)



Down View



FRONT VIEW
(With Mirror)



FRONT VIEW
(OUTSIDE)



SIDE VIEW

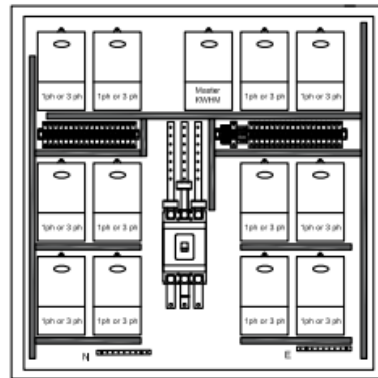
Enclosure for meters grouping V8



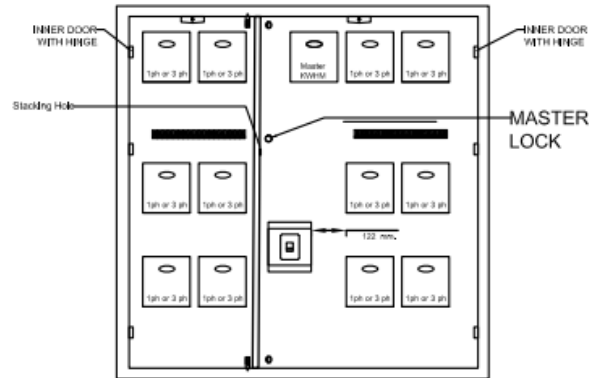
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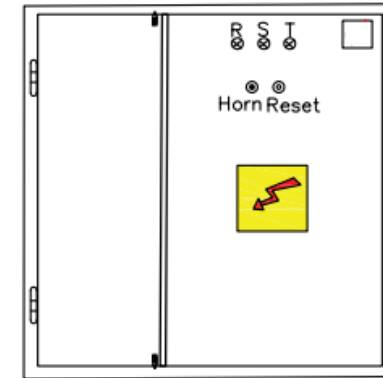
Date: 3-08-2021



FRONT VIEW
(INSIDE)



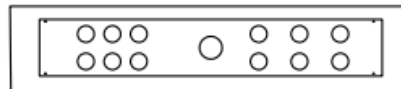
FRONT VIEW
(With Mirror)



FRONT VIEW
(OUTSIDE)



SIDE VIEW



Down View

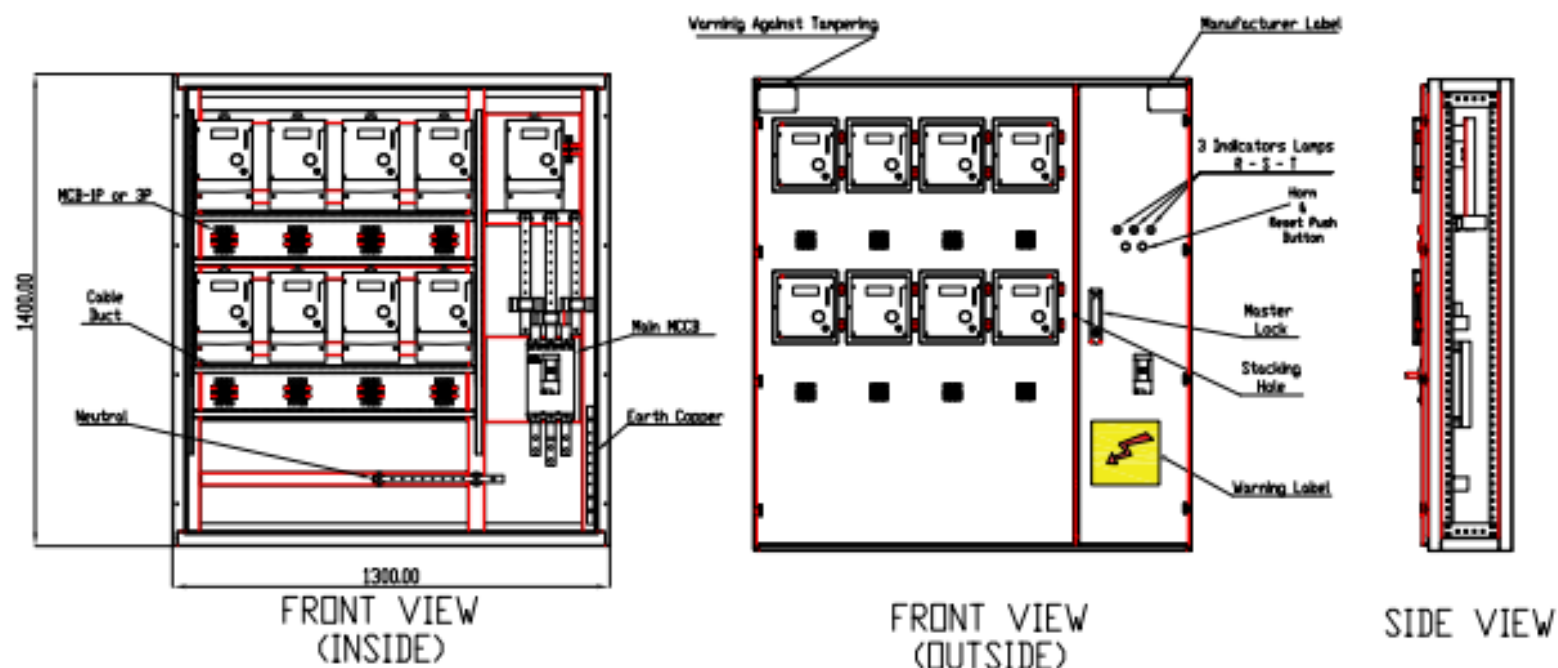
Enclosure for meters grouping V12

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14.Drawings-2

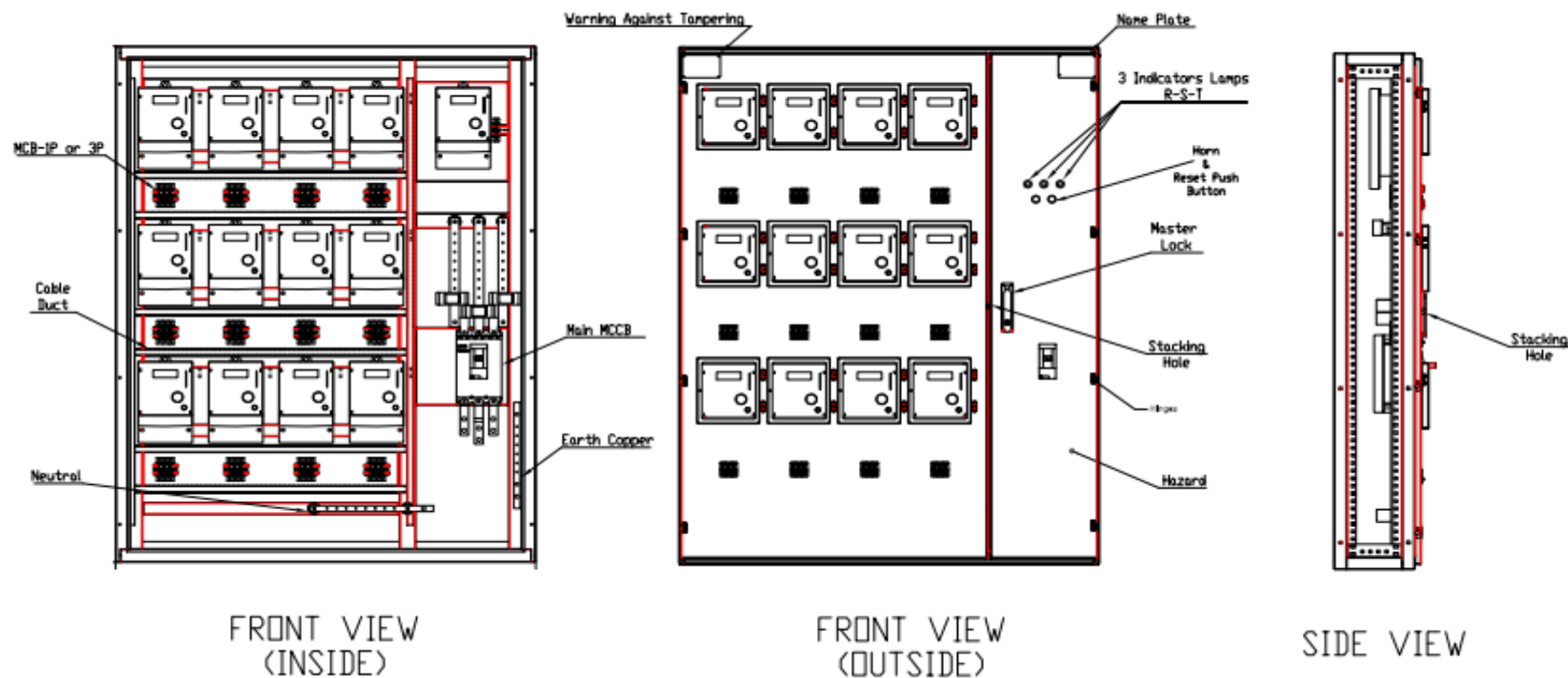


Enclosure for meters grouping V8

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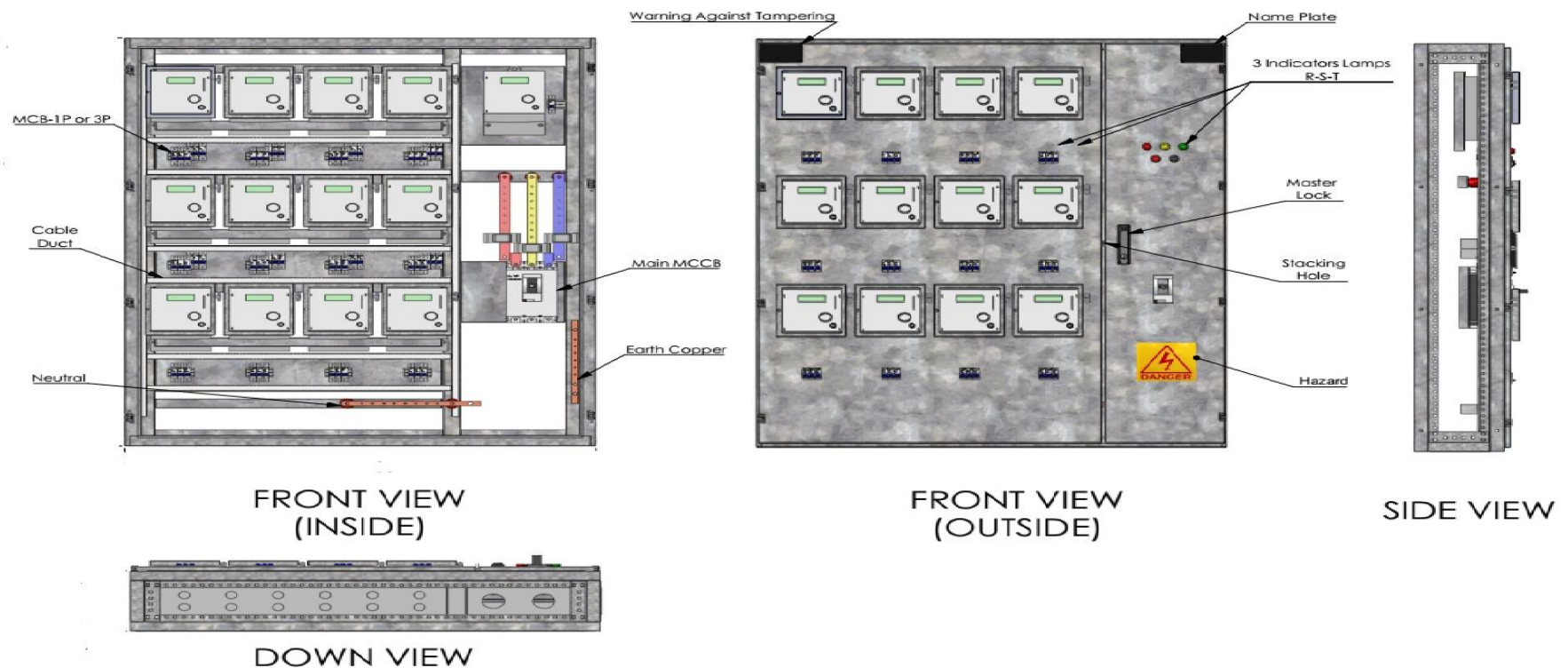
Enclosure for meters grouping V12



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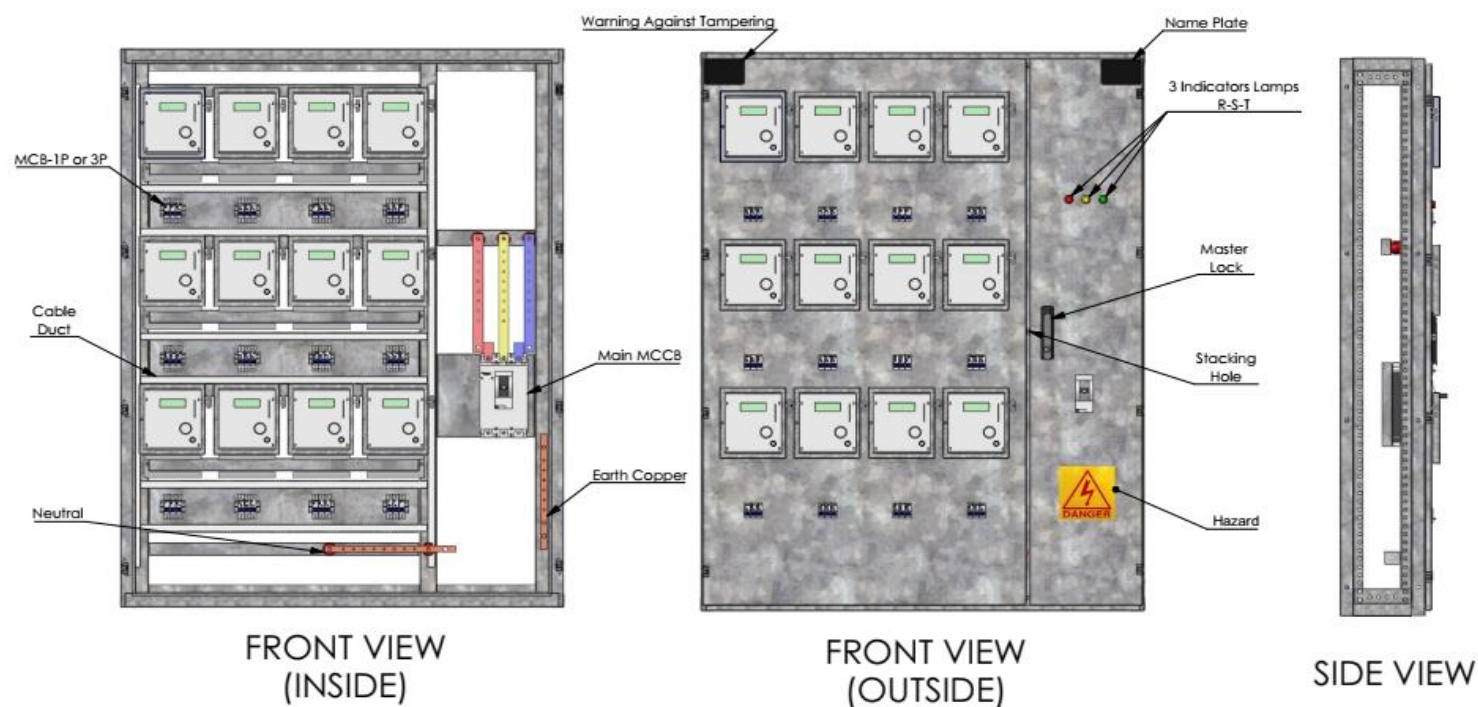


3D Dimensions of Enclosure meters grouping V12 with master meter

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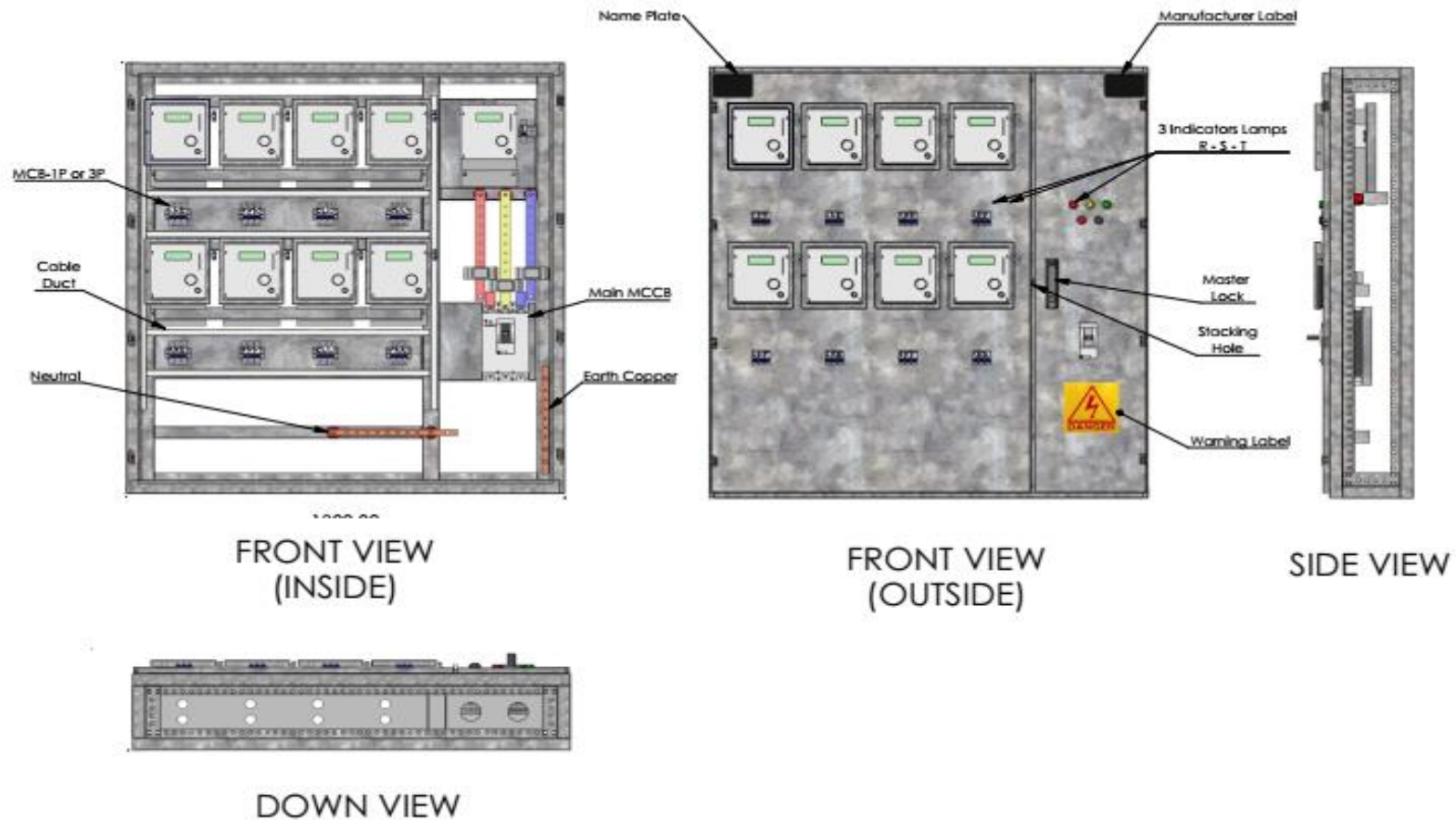


3D Dimensions of Enclosure meters grouping V12 without master meter

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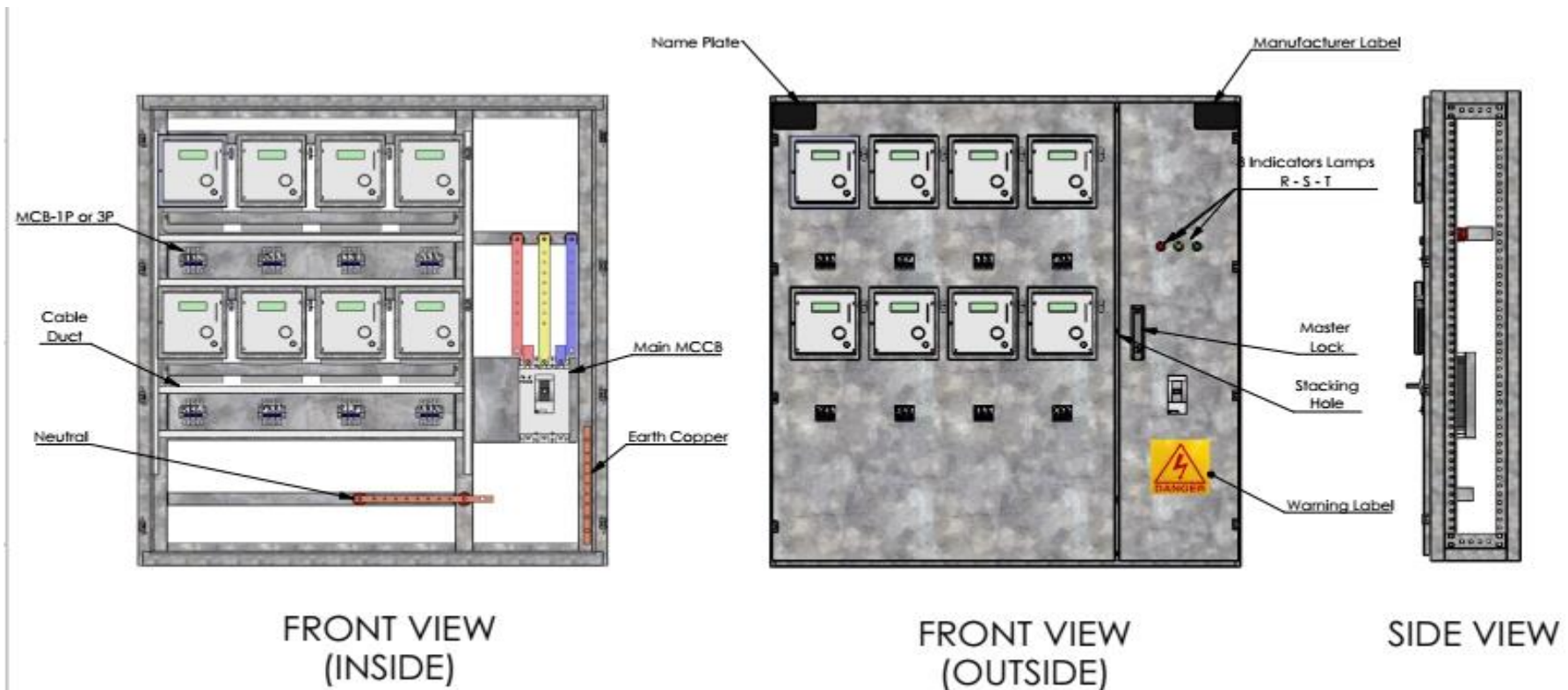
3D Dimensions of Enclosure meters grouping V8 with master meter



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3D Dimensions of Enclosure meters grouping V8 without master meter